

# AI-discovered candidate sequence factor at sgRNA position 18

Multi-model attribution and cell-line-dependent evidence from the DeepCRISPR case study

Computational discovery summary for supervisor discussion · Status: computational discovery completed; experimental validation proposed (计算发现已完成, 实验验证方案待评估)

## 1. What we found

Multiple model classes consistently identify the PAM-proximal region (positions 17–20) as a high-attribution region, with position 18 repeatedly emerging as a peak position.

多模型归因一致指向 PAM 邻近区域 (17–20 位), 第 18 位在多个模型与多个细胞系中重复成为峰值位置。

Attribution reflects model-derived importance and does not establish causality.

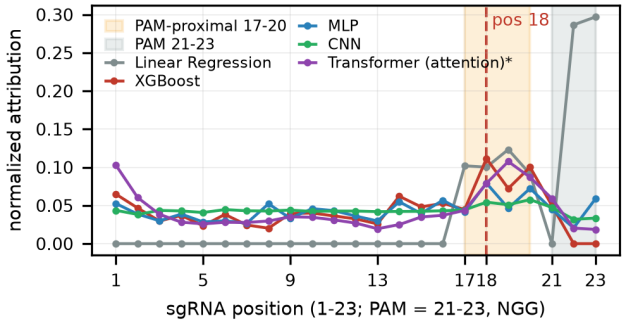


Figure 1. Normalized position attribution of five model classes (mean over cell lines; single-cell-line split). \*Transformer = attention weight, not gradient/tree attribution. Red dashed: position 18; orange: 17–20; grey: PAM 21–23.

## 2. Independent evidence

| Evidence            | Result (from artifacts)                                           |
|---------------------|-------------------------------------------------------------------|
| PAM-proximal region | highest-attribution region for CNN, MLP, XGBoost, Transformer     |
| Position 18         | recurrent peak in HCT116 and HeLa across models                   |
| Profile correlation | mean pairwise Spearman 0.28–0.61 (23 positions)                   |
| Top-3 overlap       | 0.30–0.53 of top-3 positions shared between models                |
| Model classes       | attention peaks at positions 1–2 in HCT116/HeLa (supporting only) |

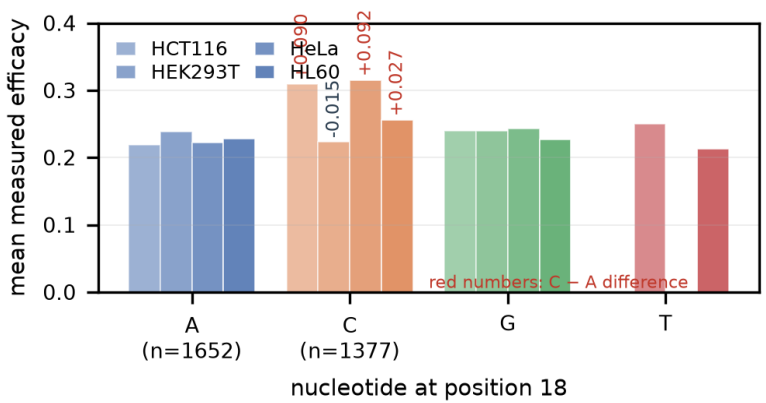


Figure 2. Mean measured editing efficiency by the nucleotide at position 18 (n = 495–3 278 per bar); numbers: C – A difference.

The position-18 nucleotide effect is context-dependent rather than universal.

第 18 位碱基与编辑效率的关系具有明显细胞背景依赖性 (HCT116/HeLa 中 C 高于 A, HL60 差异缩小, HEK293T 方向反转), 更适合作为 context-dependent candidate factor.

## 3. Current interpretation

Position 18 is a **candidate sequence-associated factor** whose predictive importance is repeatedly identified by multiple model classes, while its base-specific efficacy association varies across cell-line contexts. 第 18 位目前可定义为一个候选序列相关因素。 **This is an association / model-attribution finding, not a causal conclusion.**

Why this candidate? 01 PAM-proximal 17–20 is the highest-attribution region for CNN, MLP, XGBoost and Transformer; 02 position 18 is the modal peak in HCT116 and HeLa; 03 measured C – A difference HCT116 +0.090, HeLa +0.092, HL60 +0.027, HEK293T -0.015 (方向随细胞系改变)。

Evidence chain: multi-model attribution → PAM-proximal region → position 18 repeatedly prioritized → base-specific efficacy differences → cell-line-dependent pattern → candidate sequence factor.

## 4. Minimal experimental validation (proposed, not yet performed)

Test whether perturbing the model-prioritized nucleotide at position 18 reproducibly changes Cas9 editing efficiency.

WT: position 18 = original C

⊢ C→A ⊢ C→G ⊢ C→T

Keep the remaining sgRNA sequence unchanged.

Compare measured editing efficiency of each variant with WT.

Controls: WT + variants + negative control + reference/positive control.

Freeze the prediction before the experiment (prospective test): AI prediction (positional importance from ISM / IG / SHAP) → freeze hypothesis (position and variant set) → wet-lab experiment (WT vs variants, efficiency readout) → compare prediction with observation.

Directional mutation prediction is currently unavailable: all CNN ISM / IG / SHAP / attention outputs are unsigned magnitudes (only linear coefficients carry a sign, and they diverged in 56/192 runs). Only positional importance is supported, so the pre-registered hypothesis specifies the position and variant set, not a predicted direction.

Limitations: computational/observational evidence only; position and nucleotide-specific effects are not fully separable; cell-line differences may reflect sequence composition or cohort differences (position-18 T occurs mainly in HEK293T/HL60); wet-lab validation not yet performed.

Sources: attribution\_summary.csv, position18\_efficacy\_by\_base.csv, cross\_model\_position\_consistency.csv, cellline\_effects.csv, evidence\_matrix.csv; PAM 21–23 (NGG) verified from raw sequences. Status: computational discovery completed; validation proposed (计算发现已完成) Contact: \_\_\_\_\_